

**MCA (2YP) - 2nd Sem.
(CBEGS) (Batch 2024-26)
(2525)**

Paper: CSL-4040: Internet of Things

Time allowed: 3 hrs.

Max. Marks: 70

Note: Attempt five questions, selecting at least one question from each Section. The fifth question may be attempted from any Section. All questions carry equal marks

SECTION- A

- 1.Explain different types of Communication Protocols and their roles. (14 Marks)**
- 2.Briefly discuss various types of Sensors and its Applications. (14 Marks)**

SECTION-B

- 3.**
- (a)What do you mean by M2M Communication ? (7 Marks)**
- (b)Explain Design and Operation of Sensor Networks. (7 Marks)**
- 4.Write Short Note on:**
- (a)Interoperability in IOT (7 Marks)**
- (b)Power Optimization and Security (7 Marks)**

SECTION-C

- 5.Explain how Emerging IOT Technologies are playing crucial role in Smart Cities. (14 Marks)**
- 6.Write steps how Raspberry Pi is being Setup.Also write code of GPIO to blink an LED (7,7 Marks)**

SECTION-D

- 7.What is the role of IOT in Smart Grid Energy Optimization ? (14 Marks)**
- 8.Explain initial problem faced in Agricultural process.Why was IOT chosen as a solution for this problem. (6,8Marks)**

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